ChE 312 Final Exam example questions.

100 points, 3 problems, 5 pages.

State assumptions; be specific about units; use the right number of significant digits; explain your answers concisely. If you are short of time, be sure at least to explain how to determine the answer. NEVER ERASE! NEVER ERASE! NEVER ERASE! Use ink so you'll NEVER ERASE!

Problem 1 – Short questions (45 points).

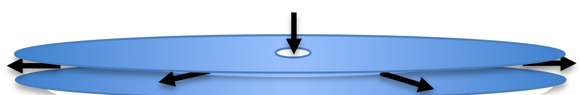
(1a, 3 points) In a hot, closed room in the summer, not a breath of air is moving. Suddenly, you smell the faint aroma of perfume. Did it reach you by diffusion or convection?

Diffusion.

(1b, 3 points) Downwind of the chicken farm, the smell of ammonia is intense; upwind, you can't smell any ammonia. What is the dominant mode of mass transfer for ammonia here, diffusion or convection?

Convective mass transfer (convection).

(1c, 4 points) Gas is injected into the center of a gap between two large, flat disks, and it uniformly flows radially outward.



Modeling the flow as steady-state with reaction and one-dimensional radial convection and diffusion, how would you simplify the following "general continuity equation for species *i*"?

$$\frac{\partial c_i}{\partial t}+\left(\upsilon_r\frac{\partial c_i}{\partial r}+\upsilon_\theta\frac{1}{r}\frac{\partial c_i}{\partial \theta}+\upsilon_z\frac{\partial c_i}{\partial z}\right)=D_i\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial c_i}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 c_i}{\partial \theta^2}+\frac{\partial^2 c_i}{\partial z^2}\right]+R_i$$

$$\cancel{\frac{\partial c_i}{\partial t}}+\left(\upsilon_r\frac{\partial c_i}{\partial r}+\cancel{\upsilon_\theta\frac{1}{r}\frac{\partial c_i}{\partial \theta}}+\cancel{\upsilon_z\frac{\partial c_i}{\partial z}}\right)=D_i\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial c_i}{\partial r}\right)+\cancel{\frac{1}{r^2}\frac{\partial^2 c_i}{\partial \theta^2}}+\cancel{\frac{\partial^2 c_i}{\partial z^2}}\right]+R_i$$

(1e, 5 points) What dimensionless number or relationship compares a substance's convective mass-transfer coefficient to its diffusivity, as in mass-transfer-coefficient correlations? Write the relationship.

The Sherwood number, N_{Sh} *or* $Sh = kd/D_A$.

(1d., 5 points) What dimensionless number or relationship compares the external convective mass-transfer coefficient of a substance to its diffusivity inside an object, as in unsteady-state mass transfer?

The Biot number, $Bi_m = kd/D_{AB}$, or $1/Bi_m = D_{AB}/k_c x_1$

[where k is mass transfer coefficient, d is characteristic length, and D_{AB} is mass diffusivity.]

(1e., 8 points) An aqueous solution of urea is in laminar flow (N_{Re}=100) against a single sphere suspended in the liquid. To calculate the flux of urea from the flowing liquid into an initially urea-free sphere, you will need a convective mass-transfer coefficient. What is a suitable correlation for obtaining it?

$N_{Sh} = k_c' D_{sphere}/D_{AB} = 2 + 0.95\, N_{Re}^{0.50}\, N_{Re}^{1/3}$ [Eq. 7.3-34]

(1f., 8 points) Obtaining a convective mass-transfer coefficient for liquid flow in a packed bed of spherical beads is more complicated than the single-sphere case of 1e, so you typically must estimate the mass-transfer coefficient by a correlation. Give an equation showing how the *J*-factor is related to the mass-transfer coefficient, and give a suitable correlation (supposing $N_{Re}\approx$100) for obtaining it.

Define $J_D = \frac{k'_c}{v_{ave}}(N_{Sc})^{2/3}$ [Eq. 7.3-13]

Correlation for liquid in a bed of spheres: $J_D = \frac{0.250}{\varepsilon} N_{Re}^{-0.31}$ [Eq. 7.3-38]

(1g., 9 points) Countercurrent gas-liquid flow in a packed bed of spherical beads is more complicated still, so you typically must obtain the mass-transfer coefficient by a correlation for H_L or H_G. Give an equation showing how H_L is related to the mass-transfer coefficient, and give a suitable correlation for obtaining it.

Relation: $H_L(m) = \frac{L}{k'_x aS} = \frac{L}{k_x(1-x)_{iM}aS}$ [Eq. 10.6-40]

For liquids in countercurrent gas-liquid flow: $H_L = \left(\frac{0.357}{f_p}\right)\left(\frac{N_{Sc}}{372}\right)^{0.5}\left(\frac{G_x/\mu}{\left(\frac{6.782}{0.8937\cdot 10^{-3}}\right)}\right)^{-0.5}$ [Eq. 10.8.2]

Problem 2 (10 points).
(2a, 5 pts) What is the difference between mass-transfer coefficients k and k'?

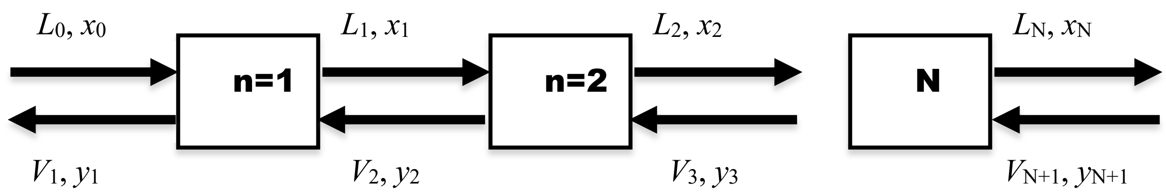
The coefficient k (without a prime) refers to the situation of **A** through non-diffusing **B**, while k' describes equimolar counterdiffusion of **A** through counterdiffusing **B**.

(2b, 5 pts) For mass transfer across an interface, convective mass-transfer resistance may be significant on either or both sides. For example, at a gas-liquid interface, $y_{\mathrm{A,interface}}$ may not be the same as $y_{\mathrm{A,bulk}}$, and $x_{\mathrm{A,interface}}$ may not be the same as $x_{\mathrm{A,bulk}}$.

How are $y_{\mathrm{A,interface}}$ and $x_{\mathrm{A,interface}}$ related to each other at steady state?

They are in equilibrium (relation such as Henry's Law or Raoult's Law).

Problem 3 (45 points).
Consider countercurrent liquid-liquid separation using equilibrium contacting stages. A feed liquid containing the solute (input rate L_0 and solute mole fraction x_0) contacts another immiscible liquid (flow and solute mole fraction labeled V_2, y_2) in the first equilibrium stage, n=1. The equilibrium relation for solute is $(\gamma_y y)_n = (\gamma_x x)_n$ (assume you have equations for finding γ_y and γ_x).



We can analyze this staged equilibrium separation with the same general approach that was used for aequilibrium-stage absorption, stripping, and distillation.
(3a, 5 points each) Write the steady-state solute material balance around stage 1.

Around stage 1, the component balance needed to construct an operating line is:
In = Out
$L_0 x_0 + V_{N+1} y_{N+1} = L_1 x_1 + V_1 y_1$ [$= Mx_M$, where $M = L_0 + V_{N+1} = L_1 + V_1$]
[Need overall balance and two of the three species balances to specify composition.]

(3b, 5 points) Using $n=N$ for the last stage, fed countercurrently by V_{N+1}, y_{N+1}, write the steady-state solute material balance for the overall system (stages 1 through N).

Around stage 1, the component balance needed to construct an operating line is:
In = Out
$L_0 x_0 + V_{N+1} y_{N+1} = L_1 x_1 + V_1 y_1$ [$= Mx_M$, where $M = L_0 + V_{N+1} = L_1 + V_1$]

(3b, 5 points) Write the material balance around the set of stages 1 through n and express it as an "operating-line equation"; that is, as $y_{n+1} = f(x_n)$.

Dilute, so balance around stages 1 through n: $L_0 x_0 + V_{n+1} y_{n+1} = L_n x_n + V_1 y_1$
Written in operating-line form: $y_{n+1} = (L_n / V_{n+1})\, x_n + [(V_1 y_1 - L_0 x_0) / V_{n+1}]$

(3c, 20 pts) Describe the steps you would use to calculate the number of stages to achieve mole fraction y_1 if you are given a solute mole fraction specification y_1? What assumptions are necessary for L and V?

On a ternary equilibrium diagram, graphically find an operating line that goes through a tie line.
The mathematical equivalent is to find an operating line $y_{N+1} = f(x_N)$ where y_{N+1} and x_N are in equilibrium.

(3d, 10 pts) You are given the conditions for a feed stream I (L_0, x_0) and solvent (V_{N+1}, y_{N+1}) and a desired mole fraction for the solute-depleted raffinate (x_N). Assume the solute is dilute; that is, L and V are ~constant. Explain the sequence of steps you would take to determine the required number of equilibrium stages using the mass balances and equilibrium relation above, analogously to stagewise calculations for distillation or absorption. *[Do not use Kremser equation or rely on having a graph of equilibrium data.]*

(1) Assume L and V are constant throughout (at least initially).
(2) Solve an overall material balance to get the mole fraction of solute in the exiting solvent, y_1.
(3) Use y_1 with the equilibrium relation to calculate x_1: $\gamma_x x_1 = \gamma_y y_1$
(4) Use the operating-line equation (material balance) and x_0 to calculate y_2.
(5) Continue alternating between the equilibrium relation and the operating-line equation until y_{n+1} reaches y_{N+1} . Then N is the number of stages required.

Alternatively, start from the raffinate product concentration x_N and the equilibrium relation to calculate y_N; then use the material balance to compute x_{N-1} and so on. Stop when x_0 and y_1 are reached.

[The Δ method recognizes that is L_n - V_{n+1} is a constant (=Δ), but this is equivalent.]

(3d, 10 pts) Describe the stagewise analysis steps used in the McCabe-Thiele algorithm (binary mixture, total condenser, partial reboiler, feed quality q), assuming known feed and distillate compositions. Compare and contrast them to the steps used in 3c.

Compare step by step:
(1) Similarly, the McCabe-Thiele algorithm assumes L and V are constant (or their ratio is) within the enriching or stripping section; constant molar overflow.
(2) Similarly, solve an overall material balance to get the mole fraction of species A in the bottoms.
(3) Similarly, use distillate mole fraction y_1 with the equilibrium relation to calculate x_1: $y_1 P = \gamma x_1 \mathrm{P}_{vap,i}$
(4) Similarly, use the operating-line equation (material balance) and x_0 to calculate y_2.
(5) Similarly, continue alternating between the equilibrium relation and the operating-line equation until (differently) the feed stage is reached. That will be determined by the operating-line equation intersecting with the q-line (feed-stage energy balance). Then switch to new operating line (and new L/V), following the same sequence until y_{n+1} reaches y_{N+1} . Then N is the number of stages required.